Data Quality Plan

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1. Introduction

After identifying issues in the refined dataset presented in the Data Quality Report, this Data Quality Plan summarizes the actions taken for each feature to address these issues, along with the reason why each action was chosen over other options.

2. Summary Table

This table summarizes all the issues encountered in cal-wildfires-23224143.csv and the corresponding solutions. It includes data type discrepancies, which were addressed in the Preliminary Data Processing section, as well as issues like missing values, which were identified but not resolved in the Data Quality Report.

Note: For features with data type issues, their data types were corrected before identifying other data quality issues. The Data Type mentioned below is the data type after conversion.

No	Feature	Data Type	Data Quality Issue	Handling Strategy
1	OBJECTID	object	Unique identifier not to be used as feature in data preparation	Keep to as the data identifier, but do not use for data preparation
2	* Damage	category	None	No action taken
3	* Street Number	category	1. Incorrect Data Type (float64) 2. Missing Values (3.50%) 3. Invalid Data (Contain 0) 4. High Cardinality (5578 cardinality)	Drop the column
4	* Street Name	category	1. Missing Values (4.43%) 2. Invalid Data (Includes only numbers) 3. High Cardinality (3956 cardinality)	Drop the column
5	* Street Type (e.g. road, drive, lane, etc.)	category	1. Missing Values (10.85%) 2. Inconsistent Data Formatting (“Hwy” vs. “Highway”) 3. Long-Tail Categories	1. Standardize categories 2. Impute missing and invalid values using "Unknown" and “Site Address (parcel)” 3. Merge categories below 1% into "Other"
6	* City	category	1. Missing Values (23.90%) 2. Inconsistent Data Formatting (trailing space, case inconsistency) 3. Long-Tail Categories	1. Standardize categories 1. Impute missing and invalid values using "Unknown"

No	Feature	Data Type	Data Quality Issue	Handling Strategy
				2. Merge categories below 0.1% into "Other"
7	State	category	Constant column(cardinality = 1)	Drop the column
8	Zip Code	category	1. Missing Values (38.92%) 2. Invalid Data (Contain 0, 19.07%)	1. Impute missing and invalid values using "Unknown" 2. impute “Unknown” using “Site Address (parcel)” and a Python package
9	* CAL FIRE Unit	category	1. Inconsistent Data Formatting (“MVU” vs. “SDU”) 2. Long-Tail Categories	Standardize categories
10	County	category	Missing Values (0.03%)	Manually retrieving missing data
11	Community	category	1. Missing Values (58.15%) 2. Inconsistent Data Formatting (“Paradise” vs. “Paradise “) 3. Long-Tail Categories	Drop the column
12	Incident Start Date	datetime	Inconsistent Data Formatting (Contain timestamp)	Save the data as date
13	Hazard Type	category	Constant column(cardinality = 1)	Drop the column
14	Structure Defense Actions Taken	category	Missing Values (72.87%)	Drop the column
15	* Structure Type	category	1. Inconsistent Data Formatting (typo for “Famliy”) 2. Long-Tail Categories	1. Standardize categories 2. Merge categories below 1% with similar ones
16	Structure Category	category	1. Inconsistent Data Formatting (incorrect category) 2. Long-Tail Categories	1. Standardize categories 2. Merge categories below 1% with similar ones
17	# Units in Structure (if multi unit)	integer	1. Missing Values (69.63%) 2. Skewed Distribution & Outliers (90% of non-missing values are 0)	Drop the column
18	* Roof Construction	category	1. Missing Values (0.49%) 2. Invalid Data (Contain blank value, 3.29%)	1. Impute missing and invalid values using "Unknown" 2. Merge categories below 1% into "Other"

No	Feature	Data Type	Data Quality Issue	Handling Strategy
19	* Eaves	category	1. Missing Values (0.96%) 2. Invalid Data (Contain blank value, 3.29%) 3. Mixed Granularity	1. Impute missing and invalid values using "Unknown" 2. Standardize categories 3. No changes were made to the granularity
20	* Vent Screen	category	1. Missing Values (0.96%) 2. Invalid Data (Contain blank value, 3.29%) 3. Mixed Granularity	1. Impute missing and invalid values using "Unknown" 2. Expanded into three distinct columns
21	* Exterior Siding	category	1. Missing Values (0.75%) 2. Invalid Data (Contain blank value, 3.29%) 3. Inconsistent Data Formatting (“Stucco Brick Cement” vs. “Stucco/Brick/Cement “)	1. Impute missing and invalid values using "Unknown" 2. Standardize categories
22	* Window Pane	category	1. Missing Values (0.75%) 2. Invalid Data (Contain blank value, 3.29%)	1. Impute missing and invalid values using "Unknown" 2. No changes were made to the granularity
23	* Deck/Porch On Grade	category	1. Invalid Data (Contain blank value, 16.61%) 2. Mixed Granularity 3. Long-Tail Categories	1. Impute missing and invalid values using "Unknown" 2. No changes were made to the granularity 3. Merge categories below 1% with similar ones to form a new category
24	* Deck/Porch Elevated	category	1. Invalid Data (Contain blank value, 16.62%) 2. Mixed Granularity 3. Long-Tail Categories	1. Impute missing and invalid values using "Unknown" 2. No changes were made to the granularity 3. Merge categories below 1% with similar ones to form a new category
25	* Patio Cover/Carport Attached to Structure	category	1. Invalid Data (Contain blank value, 16.61%) 2. Mixed Granularity	1. Impute missing and invalid values using "Unknown" 2. No changes were made to the granularity

No	Feature	Data Type	Data Quality Issue	Handling Strategy
26	* Fence Attached to Structure	category	1. Missing Values (16.61%) 2. Mixed Granularity	1. Impute missing and invalid values using "Unknown" 2. No changes were made to the granularity
27	Distance - Propane Tank to Structure	category	Missing Values (81.86%)	Drop the column
28	Distance - Residence to Utility/Misc Structure > 120 SQFT	category	1. Missing Values (85.41%) 2. Invalid Data (Contain blank value)	Drop the column
29	Assessed Improved Value (parcel)	integer	1. Missing Values (4.83%) 2. Skewed Distributions & Outliers(a significant number of extreme values)	1. Impute missing values using the median 2. Clamp outliers above the 99.5th percentile to reduce extreme values
30	Year Built (parcel)	integer	1. Missing Values (24.26%) 2. Invalid Data (758 records of 0)	1. Impute missing and invalid values using the median
31	Site Address (parcel)	category	1. Missing Values (3.55%) 2. High cardinality (8508 cardinality)	Drop the column
32	Latitude	float	None	No action taken
33	Longitude	float	None	No action taken

3. Detailed Steps

Only features with identified issues from the Data Quality Report are listeed below, along with all their associated issues, the detailed solutions applied, and the actions taken to address them. Steps such as removing constant features and converting data types were covered in the Preliminary Data Processing section of the Data Quality Report. Therefore, they will not be repeated here but can still be found in the summary table above.

* Street Number

- Issue
    - Missing Values (3.50%)
    - Invalid Data (Contain 0)
    - High Cardinality (5578 cardinality)
- Solution

    - **Drop the column:** The cardinality is too high, with values being mostly unique to each record, contributing little to meaningful predictions. Therefore, this feature has been removed.

* Street Name

- Issue
    - Missing Values (4.43%)
    - Invalid Data (Includes only numbers)
    - High Cardinality (3956 cardinality)
- Solution
    - **Drop the column:** The cardinality is too high, with values being mostly unique to each record, contributing little to meaningful predictions. Therefore, this feature has been removed.

* Street Type (e.g. road, drive, lane, etc.)

- Issue
    - Missing Values (10.85%)
    - Inconsistent Data Formatting (“Hwy” vs. “Highway”)
    - Long-Tail Categories
- Solution
    - **Standardize categories:** Replace “Hwy” with “Highway” to ensure consistency in formatting.
    - **Impute missing and invalid values using "Unknown":** Since there is limited information available to impute the missing and blank values, and "Unknown" is an already existing category, it is chosen to maintain consistency while accurately reflecting the data situation.
    - **Impute “Unknown” using “Site Address (parcel)”:** Since this feature helps standardize street classification, reducing the number of “Unknown” values is beneficial. To achieve this, after missing and invalid values were first imputed with “Unknown,” Street Type was extracted from the “Site Address (parcel)” feature by identifying similar words for greater accuracy. As a result, only 4.57% of the data remains marked as “Unknown."
    - **Merge categories below 1% into "Other"**: To mitigate the long-tail issue while preserving as much actual data as possible, a 1% threshold is applied. Categories below this threshold are merged into the 'Other' category.

* City

- Issue
    - Missing Values (23.90%)
    - Inconsistent Data Formatting (trailing space, case inconsistency)
    - Long-Tail Categories
- Solution
    - **Standardize categories**: Remove leading and trailing spaces and apply Title Case formatting to each category to reduce misclassification due to inconsistent formatting.
    - **Impute missing and invalid values using "Unknown":** Since there is limited information available to impute the missing and blank values, and "Unknown" is an already existing category, it is chosen to maintain consistency while accurately reflecting the data situation.

- **Merge categories below 0.1% into "Other":** Since the actual location can be derived from the “Latitude” and “Longitude” features, this feature primarily serves as a label for general area identification. Therefore, reducing its cardinality is beneficial. To mitigate the long-tail issue while preserving as much meaningful data as possible, a 0.1% threshold is applied, merging categories below this threshold into the 'Other' category. As a result, the cardinality is reduced from 217 to 87.

Zip Code

- Issue
  - Missing Values (38.92%)
  - Invalid Data (Contain 0, 19.07%)
- Solution
  - **Impute missing and invalid values using "Unknown":** Since there is limited information available to impute the missing and blank values, and "Unknown" is an already existing category, it is chosen to maintain consistency while accurately reflecting the data situation.
  - **Impute “Unknown” using “Site Address (parcel)” and a Python package:** Since ZIP codes provide a better understanding of geographic regions, reducing the number of “Unknown” values is beneficial. To achieve this, after missing and invalid ZIP codes were first imputed with “Unknown,” ZIP codes were extracted from the “Site Address (parcel)” feature for greater accuracy. During this process, incorrect ZIP codes were manually corrected, and a Python package was used to retrieve ZIP codes based on Latitude and Longitude. As a result, only 3.94% of the data remains marked as “Unknown."

* CAL FIRE Unit

- Issue
  - Inconsistent Data Formatting (“MVU” vs. “SDU”)
  - Long-Tail Categories
- Solution
  - **Standardize categories:** Convert “MVU” to “SDU” to enhance consistency.
  - **No other action taken:** Although more than half of the categories contribute less than 1% to the total records, merging categories solely based on a threshold would be inappropriate, as it could combine units covering entirely different areas. To preserve the integrity of the data, all categories are retained.

County

- Issue
  - Missing Values (0.03%)
- Solution
  - **Manually retrieving missing data:** Since only three records have missing values, latitude and longitude were used to retrieve county information from Google Maps.

Community

- Issue
  - Missing Values (58.15%)
  - Inconsistent Data Formatting (“Paradise” vs. “Paradise “)

    - Long-Tail Categories
- Solution
    - **Drop the column:** Due to a combination of long-tail categories and a high missing rate, imputing data is challenging, and the feature contributes little to meaningful predictions. Therefore, this feature has been removed.

Incident Start Date

- Issue
    - Inconsistent Data Formatting (Contain timestamp)
- Solution
    - **Save the data as date:** After splitting this feature into separate dates and timestamp components, it was found that only 6 records had different timestamps, while 99% of the records shared the same timestamp. Given this, creating a separate timestamp column is unnecessary. Therefore, this column was converted to a date format to maintain consistency with its column name.

Structure Defense Actions Taken

- Issue
    - Missing Values (72.87%)
- Solution
    - **Drop the column:** In addition to 72.87% missing values, 21.25% of the data is labeled as “Unknown”, resulting in 94.12% of the feature being uncertain data. Given the extremely limited number of meaningful data points, this feature has been removed as it contributes little to meaningful predictions.

* Structure Type

- Issue
    - Inconsistent Data Formatting (typo for “Famliy”)
    - Long-Tail Categories
- Solution
    - **Standardize categories:** Fix the typo of “Famliy” to ensure consistency in formatting.
    - **Merge categories below 1% with similar ones:** To mitigate the long-tail issue while preserving as much actual data as possible, a 1% threshold is applied. Categories below this threshold are merged with similar ones based on the “Structure Category” feature.

Structure Category

- Issue
    - Inconsistent Data Formatting (incorrect category)
    - Long-Tail Categories
- Solution
    - **Standardize categories:** Standardize the category of "Multi Family Residence Multi Story" to “Multi Family Residence” to ensure consistency in classification criteria.
    - **Merge categories below 1% with similar ones:** To mitigate the long-tail issue while preserving as much actual data as possible, a 1% threshold is applied. Categories below this

threshold are merged with similar ones based on the refined “* Structure Type” feature to ensure consistency.

Units in Structure (if multi unit)

- **Issue**
  - Missing Values (69.63%)
  - Skewed Distribution & Outliers (90% of non-missing values are 0)
- **Solution**
  - **Drop the column:** A total of 97.39% of records contain either 0 or missing values, making imputation both unreliable and uninformative. Given its limited analytical value, this column has been removed.

* Roof Construction

- Issue
  - Missing Values (0.49%)
  - Invalid Data (Contain blank value, 3.29%)
- Solution
  - **Impute missing and invalid values using "Unknown":** Since there is limited information available to impute the missing and blank values, and "Unknown" is an already existing category, it is chosen to maintain consistency while accurately reflecting the data situation.
  - **Merge categories below 1% into "Other":** To mitigate the long-tail issue while preserving as much actual data as possible, a 1% threshold is applied. Categories below this threshold are merged into the existing 'Other' category.

* Eaves

- Issue
  - Missing Values (0.96%)
  - Invalid Data (Contain blank value, 3.29%)
  - Mixed Granularity
- Solution
  - **Impute missing and invalid values using "Unknown":** Since there is limited information available to impute the missing and blank values, and "Unknown" is an already existing category, it is chosen to maintain consistency while accurately reflecting the data situation.
  - **Standardize categories:** Replace “Not Applicable” with “No Eaves” to ensure consistency in formatting and mitigate the long-tail issue.
  - **No changes were made to the granularity:** Although this feature exhibits mixed granularity, adding a separate 'Has/No' column would merely merge two existing categories without adding significant analytical value. After careful evaluation, it was determined that keeping the original structure is the most appropriate approach.

* Vent Screen

- Issue
  - Missing Values (0.96%)
  - Invalid Data (Contain blank value, 3.29%)
  - Mixed Granularity

- Solution
    - **Impute missing and invalid values using "Unknown":** Since there is limited information available to impute the missing and blank values, and "Unknown" is an already existing category, it is chosen to maintain consistency while accurately reflecting the data situation.
    - **Expanded into three distinct columns:** Since it is difficult to distinguish whether a vent is present and whether it has a screen, the columns 'Has Vent' and 'Has Vent Screen' were created by merging relevant categories. Additionally, the original column was renamed “Vent Screen Type” for better representation and clarity.

* Exterior Siding

- Issue
    - Missing Values (0.75%)
    - Invalid Data (Contain blank value, 3.29%)
    - Inconsistent Data Formatting (“Stucco Brick Cement” vs. “Stucco/Brick/Cement “)
- Solution
    - **Impute missing and invalid values using "Unknown":** Since there is limited information available to impute the missing and blank values, and "Unknown" is an already existing category, it is chosen to maintain consistency while accurately reflecting the data situation.
    - **Standardize categories:** Replace “Stucco/Brick/Cement” with “Stucco Brick Cement” to ensure consistency in formatting and classification.

* Window Pane

- Issue
    - Missing Values (0.75%)
    - Invalid Data (Contain blank value, 3.29%)
- Solution
    - **Impute missing and invalid values using "Unknown":** Since there is limited information available to impute the missing and blank values, and "Unknown" is an already existing category, it is chosen to maintain consistency while accurately reflecting the data situation.
    - **No changes were made to the granularity:** Although this feature exhibits mixed granularity, adding a separate 'Has/No' column would merely merge two existing categories without adding significant analytical value. After careful evaluation, it was determined that keeping the original structure is the most appropriate approach.

* Deck/Porch On Grade

- Issue
    - Invalid Data (Contain blank value, 16.61%)
    - Mixed Granularity
    - Long-Tail Categories
- Solution
    - **Impute missing and invalid values using "Unknown":** Since there is limited information available to impute the missing and blank values, and "Unknown" is an already existing category, it is chosen to maintain consistency while accurately reflecting the data situation.
    - **No changes were made to the granularity:** Although this feature exhibits mixed granularity, adding a separate 'Has/No' column would merely merge two existing categories without

adding significant analytical value. After careful evaluation, it was determined that keeping the original structure is the most appropriate approach.
    - **Merge categories below 1% with similar ones to form a new category:** To mitigate the long-tail issue while preserving as much actual data as possible, a 1% threshold is applied. 'Composite' is the only category below this threshold. Since composite materials often mimic wood, 'Composite' has been merged with 'Wood' to form the new category 'Wood (Includes Composite)' for better consistency.

* Deck/Porch Elevated

- Issue
    - Invalid Data (Contain blank value, 16.62%)
    - Mixed Granularity
    - Long-Tail Categories
- Solution
    - **Impute missing and invalid values using "Unknown":** Since there is limited information available to impute the missing and blank values, and "Unknown" is an already existing category, it is chosen to maintain consistency while accurately reflecting the data situation.
    - **No changes were made to the granularity:** Although this feature exhibits mixed granularity, adding a separate 'Has/No' column would merely merge two existing categories without adding significant analytical value. After careful evaluation, it was determined that keeping the original structure is the most appropriate approach.
    - **Merge categories below 1% with similar ones to form a new category:** To mitigate the long-tail issue while preserving as much actual data as possible, a 1% threshold is applied. 'Composite' is the only category below this threshold. Since composite materials often mimic wood, 'Composite' has been merged with 'Wood' to form the new category 'Wood (Includes Composite)' for better consistency

* Patio Cover/Carport Attached to Structure

- Issue
    - Invalid Data (Contain blank value, 16.61%)
    - Mixed Granularity
- Solution
    - **Impute missing and invalid values using "Unknown":** Since there is limited information available to impute the missing and blank values, and "Unknown" is an already existing category, it is chosen to maintain consistency while accurately reflecting the data situation.
    - **No changes were made to the granularity:** Although this feature exhibits mixed granularity, adding a separate 'Has/No' column would merely merge two existing categories without adding significant analytical value. After careful evaluation, it was determined that keeping the original structure is the most appropriate approach.

* Fence Attached to Structure

- Issue
    - Missing Values (16.61%)
    - Mixed Granularity
- Solution

- **Impute missing and invalid values using "Unknown":** Since there is limited information available to impute the missing and blank values, and "Unknown" is an already existing category, it is chosen to maintain consistency while accurately reflecting the data situation.
- **No changes were made to the granularity:** Although this feature exhibits mixed granularity, adding a separate 'Has/No' column would merely merge two existing categories without adding significant analytical value. After careful evaluation, it was determined that keeping the original structure is the most appropriate approach.

Distance - Propane Tank to Structure

- Issue
  - Missing Values (81.86%)
- Solution
  - **Drop the column:** In addition to 81.86% missing values, 5.03% of the data is labeled as “Unknown”, resulting in 86.9% of the feature being uncertain data. Given the extremely limited number of meaningful data points, this feature has been removed as it contributes little to meaningful predictions.

Distance - Residence to Utility/Misc Structure > 120 SQFT

- Issue
  - Missing Values (85.41%)
  - Invalid Data (Contain blank value)
- Solution
  - **Drop the column:** 85.41% missing data results in an extremely limited number of meaningful data points, this feature has been removed as it contributes little to meaningful predictions.

Assessed Improved Value (parcel):

- **Issue**
  - Missing Values (4.83%)
  - Skewed Distributions & Outliers(a significant number of extreme values)
- **Solution**
  - **Impute missing values using the median:** Since this feature is right-skewed, the mean is highly influenced by outliers, and the mode is 0, which does not provide meaningful information. Therefore, the median is chosen to avoid distortion caused by extreme values and ensure a more representative imputation.
  - **Clamp outliers above the 99.5th percentile to reduce extreme values:** To determine a suitable threshold for outlier clamping, we analyzed the Interquartile Range (IQR) and found 880 records exceed PR75 + 1.5 × IQR, indicating a significant number of extreme values. The PR99.5 is 27 IQRs above the 75th percentile, whereas PR99.9 spikes drastically to 182.4 IQRs, making it an unsuitable threshold. Based on this analysis, the PR99.5 was selected as the outlier threshold, and all values above this limit were clamped to PR99.5. A total of 48 records were adjusted.

Year Built (parcel)

- **Issue**

    - Missing Values (24.26%)
    - Invalid Data (758 records of 0)
- **Solution**
    - **Impute missing and invalid values using the median:** Impute missing and invalid values using the median: Since the median and mean are both 1965, this value is chosen to replace missing values and invalid entries recorded as 0.
- **Note**
    - Although 761 records fall below the PR25 - 1.5 × IQR threshold, the minimum value is only 2.36 IQRs below the PR25, indicating no severe anomalies. Additionally, the data ranges between 1856 and 2022, with no significant issues requiring correction. No further action is needed.

Site Address (parcel)

- Issue
    - Missing Values (3.55%)
    - High cardinality (8508 cardinality)
- Solution
    - **Drop the column:** The cardinality is too high, with values being mostly unique to each record, contributing little to meaningful predictions. Therefore, after using this feature to help reduce “Unknown” values in “Street Type” and “ZIP Code”, this feature has been removed.